

Roger Richard Demello

Software Engineer | Nagpur, India | B.Tech 2026 | Immediate joiner | Open to relocating internationally
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PROFESSIONAL SUMMARY

Software engineer who ships end to end and owns it in production. Built and operated **3 production LLM applications** with **RAG pipelines** for **200+ users** at **99.5% uptime**, **latency cut 35%**, in **Python** and **FastAPI**. **AWS Certified**.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, SQL, Bash, YAML – data structures, algorithms, object-oriented programming
Machine Learning: Scikit-learn, PyTorch, XGBoost, LightGBM, Feature Engineering, Model Evaluation, Anomaly Detection
Generative AI (GenAI): Retrieval-Augmented Generation (RAG), LLMs, Prompt Engineering, LangChain, Agentic AI
Backend and APIs: REST APIs, Microservices, FastAPI, Flask, Node.js, Express.js, WebSockets, Webhooks, System Design
Data and Databases: PostgreSQL, MySQL, SQLite, Redis, Vector Databases, FAISS, SQLAlchemy, Prisma ORM, Pandas
Cloud, DevOps and Testing: AWS (Certified), Docker, Kubernetes-ready, CI/CD, GitHub Actions, Linux, Git, pytest, Agile

WORK EXPERIENCE

Innovun Global August 2026 – Present

AI Engineer Intern, Remote

- Develop a **multi-channel enrollment agent** for client Cruz Roja Mexicana across **3 channels** – WhatsApp, Instagram, web.
- Integrate the **WhatsApp Business Cloud API** and **Instagram Graph API** over **webhooks** behind **FastAPI REST APIs**.
- Build the **Retrieval-Augmented Generation (RAG)** pipeline – document chunking, embedding models, vector search.
- Collaborate **cross-functionally** with client stakeholders in an **Agile** remote team, authoring **technical documentation**.

AI LifeBOT

January 2026 – June 2026

AI Engineer Intern

- Built and operated **3 production LLM applications** (Python, FastAPI, PostgreSQL) for **200+ users** at **99.5% uptime**.
- **Cut response latency 35%** by profiling hot paths in an inherited codebase, owning incident response after go-live.
- Shipped **5+ features end to end 50% faster** with **pytest suites in CI**, code reviews and **telemetry and audit logging**.

CFM, RCOEM

May 2025 – July 2025

Machine Learning Research Intern

- Built a prediction system over **1,000+ patient health records** at **87% accuracy**, owning the pipeline end to end.
- Engineered **biometric features** – blood pressure, sleep duration, stress indices – from raw tabular clinical data.
- Optimized data pipelines (Python, pandas), cutting development time **30%** and raising **validation accuracy 25%**.

PROJECTS

SentinelZero | *Python, FastAPI, SQLite, WebSockets, in-memory vector search, Docker* | [GitHub](#)

- Engineered an industrial safety platform ingesting live sensor data and scoring asset risk in real time.
- Architected a **modular monolith of 6 services** – ingestion, risk engine, retrieval, simulation, coaching, auth.
- Streamed **WebSocket** telemetry over **10 assets**, scored **0-100** by z-score **anomaly detection** and rules.
- Secured the platform with **JWT authentication** and **role-based access control** over an in-process **event bus**.
- Delivered a runnable stack with **zero external services** – SQLite, in-memory vector search, offline mock.

DealSentry | *Node.js, Express.js, Prisma ORM, PostgreSQL, Supabase, OpenAI API, Puppeteer* | [GitHub](#)

- Developed a proposal compliance and risk-review platform for sales teams to analyze and approve proposals.
- Automated compliance checks with a **rules engine** and **risk scoring**, catching pricing and legal issues early.
- Implemented **role-based access control** with structured review stages and approval routing under **SLA tracking**.
- Integrated **4 enterprise systems** – Salesforce, HubSpot, Gmail and Google Drive – with upload and auto-parsing.

music-recsys | *Python, PyTorch, LightGBM, FastAPI, MLflow, Redis, Kafka, Prometheus, Docker* | [GitHub](#)

- Designed a two-stage recommender – two-tower embeddings feed **ANN retrieval** into a **LightGBM** ranker.
- Decoupled **5 subsystems** behind Protocols – cache, event bus, store, ANN index and model registry.
- Scaled from CPU-only local runs to **Docker, Kafka and Kubernetes** by flipping one config value.
- Monitored a **REST API** with **Prometheus and Grafana**, plus an **A/B simulator** for controlled rollouts.

EDUCATION

Shri Ramdeobaba College of Engineering and Management (RCOEM), Nagpur

2022 – 2026

B.Tech, Electronics & Communication Engineering

GPA: 8.90/10

Minor: **Artificial Intelligence & Machine Learning**

GPA: 9.67/10

CERTIFICATIONS AND AWARDS

AWS Certified Cloud Practitioner – Amazon Web Services (October 2025)

Hackathon Finalist – Paytm x Sarvam x Logitech AI National Hackathon

Second Place – ByteSize Sage AI National Hackathon